25901397 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMBG200 - ENVIRONMENTAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. What is liquid waste? [K] / [CO3]
2. What is an Estuary? [K] / [CO2]
3. List out any two types of biogas plants. [K] / [CO3]
4. Define gravity-based air sampling and name one device used for it. [K] / [CO1]
5. What is an Oxidation pond? [K] / [CO2]
6. What are the disadvantages of Ozone for water purification? [K] / [CO2]
7. State one environmental factor affecting hydrocarbon degradation. [K] / [CO3]
8. State what the Pelagic zone refers to. [K] / [CO2]
9. What is vertical migration? [K] / [CO2]

10. What are the different phases of composting? [U] / [CO3]

11. Explain the function of biosurfactants in oil degradation. [U] / [CO3]

12. What is Biofouling? [K] / [CO2]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Describe the mechanisms by which microbes from humans are dispersed into the air. [U] / [CO1]

14. Give a brief description of the zonation of Mangroves based on salinity. [K] / [CO2]

15. Apply the concept of phytovolatilization to remediate mercury-contaminated soil. [A] / [CO3]

16. Explain why Penicillium and Aspergillus fungi are considered good indicators of indoor air quality and dampness problems. [U] / [CO1]

17. Explain eutrophication. [U] / [CO2]

18. Suggest a suitable bioremediation strategy for oil-contaminated coastal soil and justify your choice. [A] / [CO3]

[5x4 = 20]

Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Discuss the concept of cometabolism in pesticide degradation. [U] / [CO3]

20. Write a detailed note on water pollution and its sources. [K] / [CO2]

[10x1 = 10]